49. Cyclic Redundancy Check (CRC)

49.1 Overview

The Cyclic Redundancy Check (CRC) calculator generates CRC codes to detect errors in the data. The bit order of CRC calculation results can be switched for LSB-first or MSB-first communication. Additionally, various CRC-generation polynomials are available. The snoop function allows monitoring reads from and writes to specific addresses. This function is useful in applications that require CRC code to be generated automatically in certain events, such as monitoring writes to the serial transmit buffer and reads from the serial receive buffer.

Table 49.1 lists the CRC calculator specifications and Figure 49.1 shows a block diagram.

Table 49.1 CRC calculator specifications

Item	Description	
Data size	8-bit	32-bit
Data for CRC calculation*1	CRC code generated for data in 8n-bit units (where n is a natural number)	CRC code generated for data in 32n-bit units (where n is a natural number)
CRC processor unit	Operation executed on 8 bits in parallel	Operation executed on 32 bits in parallel
CRC generating polynomial	One of three generating polynomials that is selectable: [8-bit CRC] • $X^8 + X^2 + X + 1$ (CRC-8) [16-bit CRC] • $X^{16} + X^{15} + X^2 + 1$ (CRC-16) • $X^{16} + X^{12} + X^5 + 1$ (CRC-CCITT)	One of two generating polynomials that is selectable: [32-bit CRC] • $X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$ (CRC-32) • $X^{32} + X^{28} + X^{27} + X^{26} + X^{25} + X^{23} + X^{22} + X^{20} + X^{19} + X^{18} + X^{14} + X^{13} + X^{11} + X^{10} + X^9 + X^8 + X^6 + 1$ (CRC-32C)
CRC calculation switching	The bit order of CRC calculation results can be switched for LSB-first or MSB-first communication	
Module-stop function	Module-stop state can be set to reduce power consumption	
CRC snoop	Monitor reads from and writes to a certain register address	
TrustZone Filter	Security and Privilege attribution can be set	

Note 1. This function cannot divide data used in CRC calculations. Write data in 8-bit or 32-bit units.

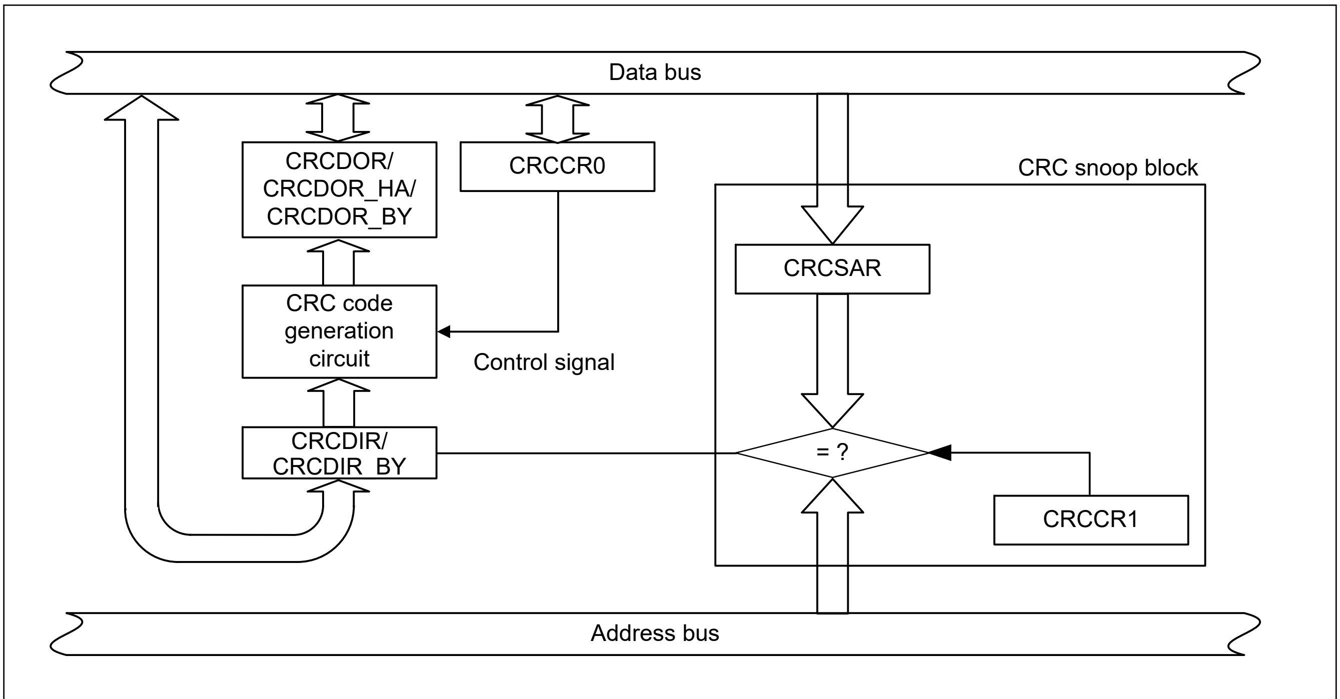

Figure 49.1 CRC calculator block diagram

49.2 Register Descriptions

49.2.1 CRCCR0 : CRC Control Register 0

Base address: CRC = 0x4031_0000
CRC_NS = 0x5031_0000

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DORCLR	LMS	—	—	—	GPS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	GPS[2:0]	CRC Generating Polynomial Switching 0 0 1: 8-bit CRC-8 ($X^8 + X^2 + X + 1$) 0 1 0: 16-bit CRC-16 ($X^{16} + X^{15} + X^2 + 1$) 0 1 1: 16-bit CRC-CCITT ($X^{16} + X^{12} + X^5 + 1$) 1 0 0: 32-bit CRC-32 ($X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$) 1 0 1: 32-bit CRC-32C ($X^{32} + X^{28} + X^{27} + X^{26} + X^{25} + X^{23} + X^{22} + X^{20} + X^{19} + X^{18} + X^{14} + X^{13} + X^{11} + X^{10} + X^9 + X^8 + X^6 + 1$) Others: No calculation is executed	R/W
5:3	—	These bits are read as 0. The write value should be 0.	R/W
6	LMS	CRC Calculation Switching 0: Generate CRC code for LSB-first communication 1: Generate CRC code for MSB-first communication	R/W
7	DORCLR	CRCDOR/CRCDOR_HA/CRCDOR_BY Register Clear 0: No effect 1: Clear the CRCDOR/CRCDOR_HA/CRCDOR_BY register	W

Note: S-TYPE3, P-TYPE3

GPS[2:0] bits (CRC Generating Polynomial Switching)

The GPS[2:0] bits select the CRC generating polynomial.

LMS bit (CRC Calculation Switching)

The LMS bit selects the bit order of generated CRC code. Transmit the lower byte of the CRC code first for LSB-first communication and the upper byte first for MSB-first communication. For details on transmitting and receiving CRC code, see section 49.3. Operation.

DORCLR bit (CRCDOR/CRCDOR_HA/CRCDOR_BY Register Clear)

Write 1 to the DORCLR bit to set the CRCDOR/CRCDOR_HA/CRCDOR_BY register to 0x00000000. This bit is read as 0. Only 1 can be written to it.

49.2.2 CRCCR1 : CRC Control Register 1

Base address: CRC = 0x4031_0000
CRC_NS = 0x5031_0000

Offset address: 0x01

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CRCSEN	CRCSWR	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	—	These bits are read as 0. The write value should be 0.	R/W
6	CRCSWR	Snoop-On-Write/Read Switch 0: Snoop-on-read 1: Snoop-on-write	R/W
7	CRCSEN	Snoop Enable 0: Disabled 1: Enabled	R/W

Note: S-TYPE3, P-TYPE3

CRCSWR bit (Snoop-On-Write/Read Switch)

The CRCSWR bit selects the direction of access in the CRC snoop function.

When this bit is set to 0 (initial value), the CRC snoop operation to read a specific register is enabled. Similarly, when this bit is set to 1, the CRC snoop operation to write a specific register is enabled.

CRCSEN bit (Snoop Enable)

When the CRCSEN bit is set to 1, the CRC snoop operation is enabled. When this bit is set to 0, the CRC snoop operation is disabled.

49.2.3 CRCDIR/CRCDIR_BY : CRC Data Input Register

Base address: CRC = 0x4031_0000
CRC_NS = 0x5031_0000

Offset address: 0x04

Bit position: 31 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	CRC input data The CRCDIR register is a 32-bit read/write register to write data for CRC-32 or CRC-32C calculation. The CRCDIR_BY (CRCDIR[31:24]) is an 8-bit read/write register to write data for CRC-8, CRC-16, or CRC-CCITT calculation.	R/W

Note: S-TYPE3, P-TYPE3

49.2.4 CRCDOR/CRCDOR_HA/CRCDOR_BY : CRC Data Output Register

Base address: CRC = 0x4031_0000
CRC_NS = 0x5031_0000

Offset address: 0x08

Bit position: 31 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	CRC output data The CRCDOR register is a 32-bit read/write register for CRC-32 or CRC-32C calculation. The CRCDOR_HA (CRCDOR[31:16]) register is a 16-bit read/write register for CRC-16 or CRC-CCITT calculation. The CRCDOR_BY (CRCDOR[31:24]) register is an 8-bit read/write register for CRC-8 calculation. Because its initial value is 0x00000000, rewrite the CRCDOR/CRCDOR_HA/ CRCDOR_BY register to perform the calculations using a value other than the initial value. Data written to the CRCDIR/CRCDIR_BY register is CRC calculated and the result is stored in the CRCDOR/CRCDOR_HA/CRCDOR_BY register. If the CRC code is calculated following the transferred data and the result is 0x00000000, there is no CRC error.	R/W

Note: S-TYPE3, P-TYPE3

49.2.5 CRCSAR : Snoop Address Register

Base address: CRC = 0x4031_0000
CRC_NS = 0x5031_0000

Offset address: 0x0C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CRCSA[13:0]													
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
13:0	CRCSA[13:0]	Register Snoop Address These bits store the TDR or RDR address in the SCI module to snoop	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

CRCSA[13:0] bits (Register Snoop Address)

The CRCSA[13:0] bits specify the lower address 14 bits of the register monitored by the CRC snoop operation.

Only the following addresses can be used for the CRCSA[13:0] bits:

[Secure region address]

- 0x4035_8004: SCI0.TDR, 0x4035_8000: SCI0.RDR
- 0x4035_8104: SCI1.TDR, 0x4035_8100: SCI1.RDR
- 0x4035_8204: SCI2.TDR, 0x4035_8200: SCI2.RDR
- 0x4035_8304: SCI3.TDR, 0x4035_8300: SCI3.RDR
- 0x4035_8404: SCI4.TDR, 0x4035_8400: SCI4.RDR
- 0x4035_8504: SCI5.TDR, 0x4035_8500: SCI5.RDR
- 0x4035_8604: SCI6.TDR, 0x4035_8600: SCI6.RDR
- 0x4035_8704: SCI7.TDR, 0x4035_8700: SCI7.RDR
- 0x4035_8804: SCI8.TDR, 0x4035_8800: SCI8.RDR
- 0x4035_8904: SCI9.TDR, 0x4035_8900: SCI9.RDR

[Non-secure region address]

- 0x5035_8004: SCI0_NS.TDR, 0x5035_8000: SCI0_NS.RDR
- 0x5035_8104: SCI1_NS.TDR, 0x5035_8100: SCI1_NS.RDR
- 0x5035_8204: SCI2_NS.TDR, 0x5035_8200: SCI2_NS.RDR
- 0x5035_8304: SCI3_NS.TDR, 0x5035_8300: SCI3_NS.RDR
- 0x5035_8404: SCI4_NS.TDR, 0x5035_8400: SCI4_NS.RDR

- 0x5035_8504: SCI5_NS.TDR, 0x5035_8500: SCI5_NS.RDR
- 0x5035_8604: SCI6_NS.TDR, 0x5035_8600: SCI6_NS.RDR
- 0x5035_8704: SCI7_NS.TDR, 0x5035_8700: SCI7_NS.RDR
- 0x5035_8804: SCI8_NS.TDR, 0x5035_8800: SCI8_NS.RDR
- 0x5035_8904: SCI9_NS.TDR, 0x5035_8900: SCI9_NS.RDR

49.3 Operation

49.3.1 Basic Operation

The CRC calculator generates CRC codes for use in LSB-first or MSB-first transfer.

The following examples show CRC code generation for input data (0xF0) using the 16-bit CRC-CCITT generating polynomial ($X^{16} + X^{12} + X^5 + 1$). In these examples, the value of the CRC Data Output Register (CRCDOR_HA) is cleared before CRC calculation.

When an 8-bit CRC (with the polynomial $X^8 + X^2 + X + 1$) is in use, the valid bits of the CRC code are obtained in CRCDOR_BY. When a 32-bit CRC is in use, the valid bits of the CRC code are obtained in CRCDOR.

Figure 49.2 and Figure 49.3 show the LSB-first and MSB-first data transmission examples respectively. Figure 49.4 and Figure 49.5 show the LSB-first and MSB-first data reception examples.

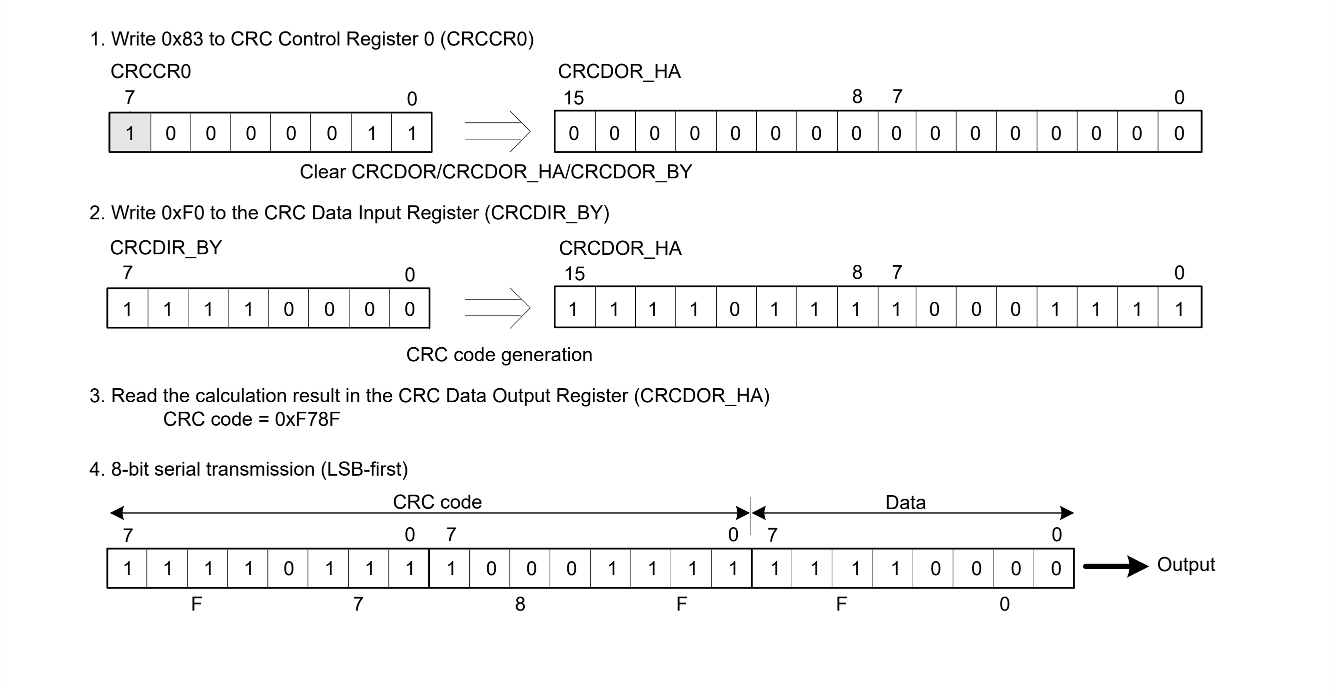

Figure 49.2 LSB-first data transmission

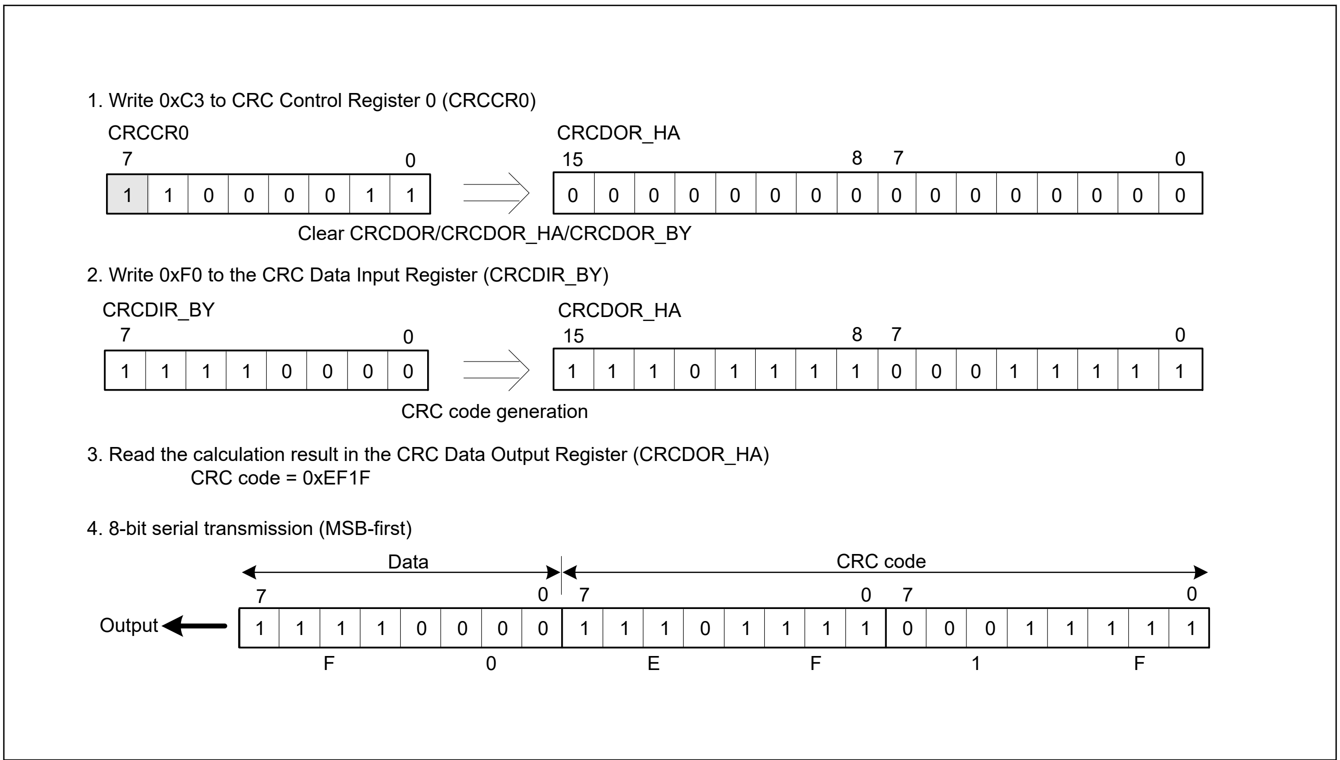

Figure 49.3 MSB-first data transmission

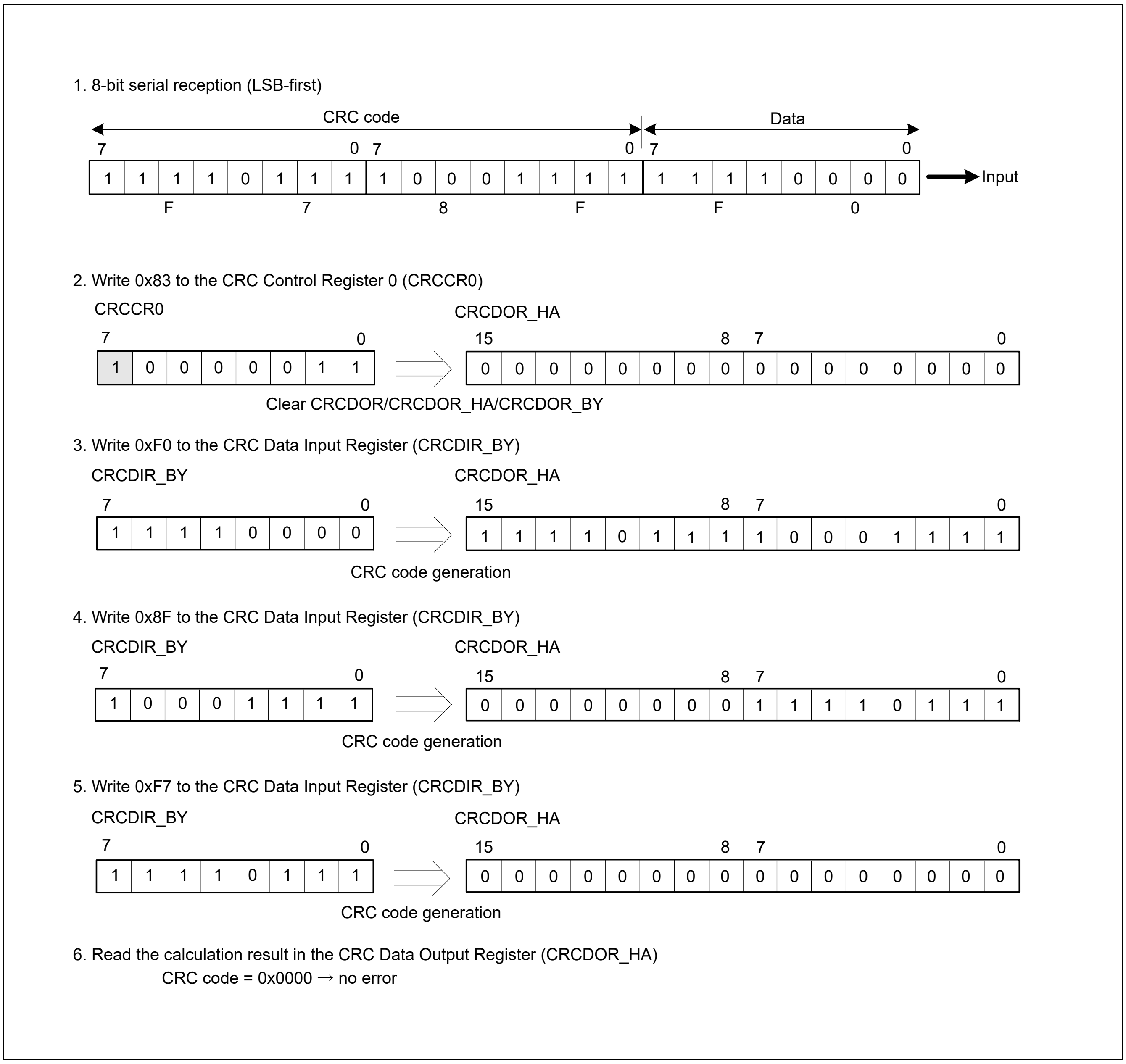

Figure 49.4 LSB-first data reception

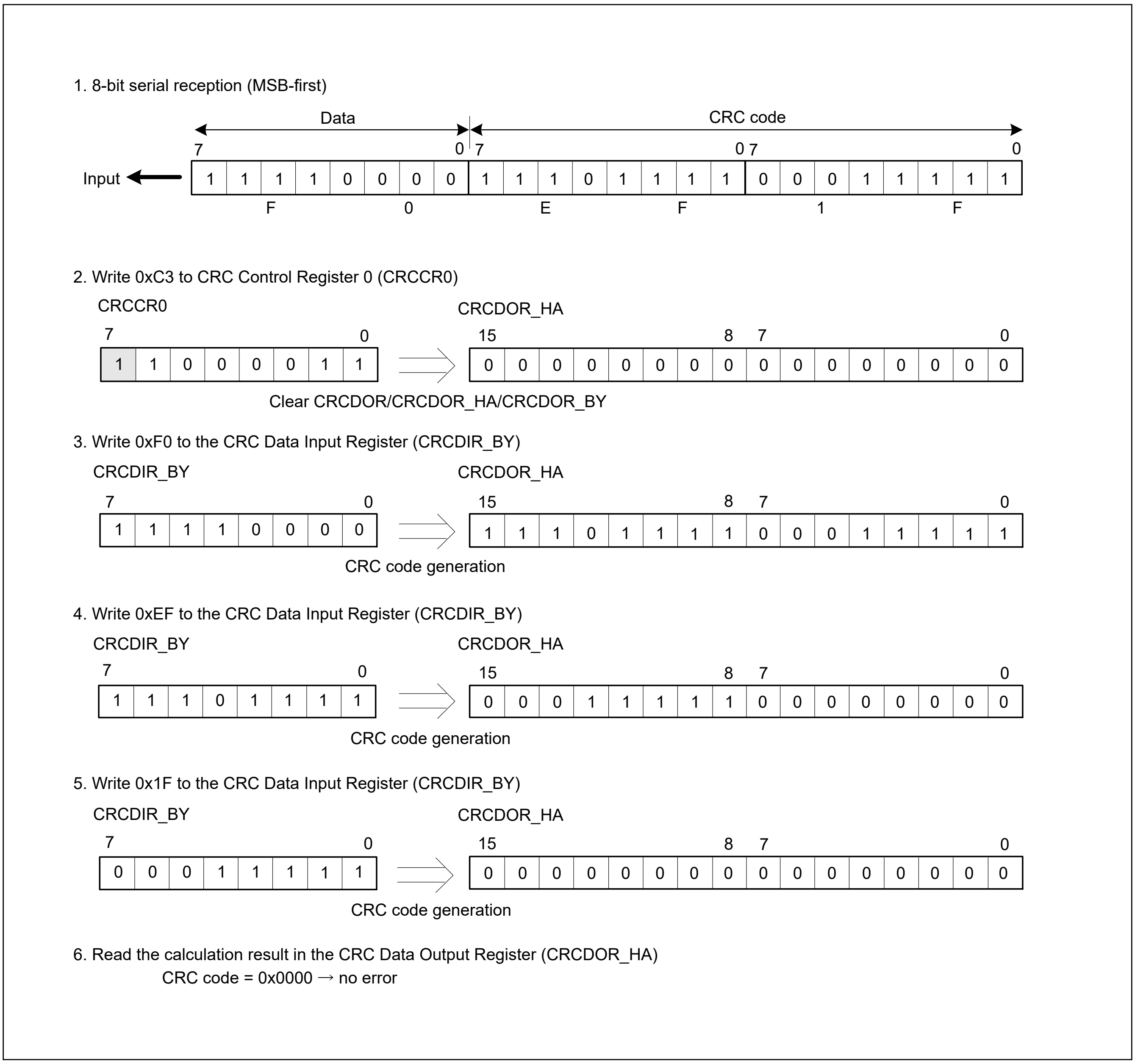

Figure 49.5 MSB-first data reception

49.3.2 CRC Snoop Function

The CRC snoop function monitors reads from and writes to a specific register and performs CRC calculation on the monitored data automatically. Because the CRC snoop function recognizes writes to and reads from a specific register address as a trigger to automatically perform CRC calculation, there is no need to write data to the CRCDIR register. All I/O registers specified in the section 49.2.5. CRCSAR : Snoop Address Register are subject to the CRC snoop. The CRC snoop is useful in monitoring writes to the SCIn.TDR (n = 0 to 9) register, and reads from the SCIn.RDR (n = 0 to 9) register.

To use this function, write the lower address 14 bits of a specific register to bits CRCSA[13] to CRCSA[0] in the CRCSAR register, and set CRCSEN bit in the CRCCR1 register to 1. Then, set the CRCSWR bit in the CRCCR1 register to 1 to enable snooping on writes to the target register, or set the CRCSWR bit in the CRCCR1 register to 0 to enable snooping on reads from the target register. It is possible that access to a target I/O register may be executed before the CRCSWR bit write completed. In this case, the data is not stored in the CRCDIR register. To avoid this issue, before accessing I/O register, read back the CRCSWR bit that was written to confirm that the write completed.

When both the CRCSEN and CRCSWR bits are set to 1, and data is written to a target register in a bus master module such as the CPU, DMAC, and DTC, the CRC calculator stores the data in the CRCDIR register and performs CRC calculation. Similarly, when the CRCSEN bit is set to 1, CRCSWR bit to 0, and data is read from a target register in a bus master module such as the CPU, DMAC, and DTC, the CRC calculator stores the data in the CRCDIR register and performs CRC calculations.

When the CRC code is generated by using CRC-8, CRC-16, and CRC-CCITT generating polynomial, the target register is accessed in 1 byte (8 bits). RDR_BY and TDRLL should be used to access RDR and TDR. Similarly, when the CRC code is generated by using CRC-32 and CRC-32C generating polynomial, the target register is accessed in 1 word (32 bits). Note that for RDR and TDR, CRC codes are generated that contain data other than RDAT and TDAT.

When CRC is marked as secure by PSARC.PSARC1 bit, the CRC snoop function is available for secure access to the specified I/O registers. When CRC is marked as non-secure by PSARC.PSARC1 bit, the CRC snoop function is available for non-secure access to the specified I/O registers.

49.4 Usage Notes

49.4.1 Settings for the Module-Stop State

The Module Stop Control Register C (MSTPCRC) can enable or disable CRC calculator operation. The CRC calculator is initially stopped after a reset. Releasing the module-stop state enables access to the registers. For details, see section 11, Low Power Mode.

49.4.2 Note on Transmission

The transmission sequence for the CRC code differs based on whether the transmission is LSB-first or MSB-first. Figure 49.6 shows an LSB-first and MSB-first data transmission.

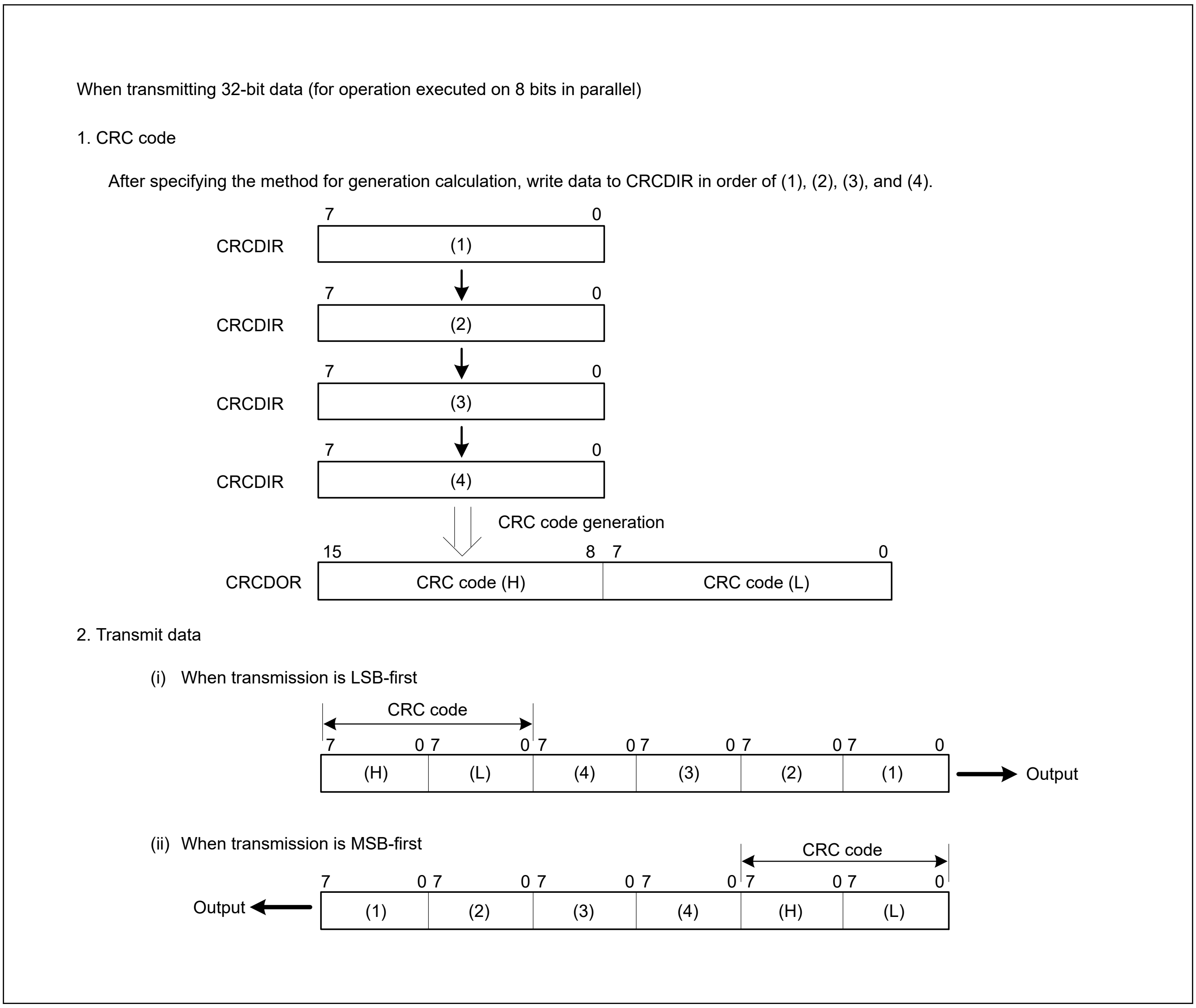

Figure 49.6 LSB-first and MSB-first data transmission